

Pedi-IKDC exhibits questionable measurement properties in a cohort of pediatric patients with ACL rupture

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Background: Pedi-IKDC is commonly used to evaluate anterior cruciate ligament (ACL) deficiency in children. However, its construct validity has not been thoroughly assessed. The aim was to examine the measurement properties of the Pediatric International Knee Documentation Committee (Pedi-IKDC) by modern test theory (MTT) models, confirmatory factor analysis (CFA), and item response theory (IRT).

Methods: The cohort consisted of all children and adolescents in Denmark ($n = 535$, age 9–16) treated with physeal-sparing ACL reconstruction 2011–2020. Patient-reported outcome measure (PROM) data were collected before surgery and at 1 year follow-up. Structural validity of Pedi-IKDC was assessed with MTT models. Reliability was reported as McDonalds coefficient omega. Responsiveness was evaluated with standardized response means.

Results: Sufficient PROM data were available for 372 patients. The original unidimensional construct did not fit CFA model expectations neither before surgery ($\chi^2 = 462.0$, $df = 163$, $p < 0.0001$; RMSEA: 0.109, CFI: 0.910, TFI: 0.895) nor at follow-up. Neither did a two-factor CFA model with “Symptoms” and “Sports activities” as individual subscales ($\chi^2 = 455.6$, $df = 162$, $p < 0.0001$) nor a bifactor model ($\chi^2 = 338.9$, $df = 143$, $p < 0.0001$), although fit indices improved with the latter (RMSEA: 0.094, CFI: 0.941, TFI: 0.922). The IRT models confirmed this pattern. The scale was responsive (SRM 1.66 (95% CI: 1.46–1.88)). Coefficient omega values were 0.866 before surgery and 0.919 at follow-up.

Conclusions: The Pedi-IKDC exhibited inadequate structural validity. Neither the original construct, a two-factor model, nor bifactor models fitted data well. We advise that data obtained by Pedi-IKDC are interpreted with caution.

KEYWORDS

anterior cruciate ligament reconstruction, construct validity, measurement properties, patient reported outcome measures, pediatrics, structural validity

1 | INTRODUCTION

Rupture of the anterior cruciate ligament (ACL) is an uncommon injury in children,^{1,2} and it affects their ability to be normally active.³ More than half of pediatric patients with ACL injury are treated by ligament reconstruction with physeal-sparing techniques. The treatment results are assessed by objective measures (like mechanical stability), functional tests (like three single-leg hop test), and the child's own experience expressed through patient-reported outcome measures (PROMs) in the form of questionnaires. Data obtained by PROMs are often regarded as the most important outcome.⁴

The most used pediatric PROM to evaluate the treatment of children with ACL deficiency is the Pediatric International Knee Documentation Committee Subjective Knee Form (Pedi-IKDC).⁵ It is a modification of a PROM developed for adults with various knee conditions, the International Knee Documentation Committee Subjective Knee Form (IKDC), and it is meant as an evaluative tool for children or adolescents with any knee injury.^{6,7}

Pedi-IKDC was developed in Boston, United States, in 2010 by asking 30 children with a knee injury to complete the adult 19-item IKDC, while cognitive interviewing methods were used to explore emerging difficulties.⁶ On this basis, a 20-item Pedi-IKDC questionnaire was created. Assessment of construct validity, reliability, and responsiveness was performed by classical test theory (CTT) methods, and it was concluded that the Pedi-IKDC is valid, reliable, and responsive.^{6,8} The measurement properties have subsequently been validated by CTT analyses with the same conclusion for American children/adolescents with ACL injury,⁹ and for children with a variety of knee conditions in Danish,¹⁰ Dutch,¹¹ and Italian¹² populations. However, there are several limitations to analyses by CTT methods, and the most reliable methods to validate the measurement properties of a PROM is through modern test theory (MTT) models.^{13–15} There is currently no published assessment of the construct validity of Pedi-IKDC by MTT methods. Thus, the level of evidence for construct validity for children with knee disorders in general is currently low for Pedi-IKDC, and there is no high-level evidence that it is an adequate measurement instrument for children with ACL deficiency.

Therefore, the aim of this study was to assess the structural validity, reliability, and responsiveness of Pedi-IKDC in a nation-wide cohort of Danish children with ACL deficiency, using the most rigorous statistical validation methods: confirmatory factor analysis (CFA) and item response theory (IRT).^{14,15}

2 | MATERIALS AND METHODS

2.1 | Patients and data collection

Since 2011, the surgical treatment of children with ACL injury in Denmark has been centralized at two centers: one at Aarhus University Hospital and one at Bispebjerg and Frederiksberg Copenhagen University Hospital. This includes all children <16 years of age. Children with open physes are reconstructed with a physeal-sparing technique. Concomitant meniscal and/or cartilage injuries were simultaneously addressed when needed, in a one-step procedure. Injuries to collateral ligaments were treated conservatively. The cohort is nation-wide, as these patients are not reconstructed in other hospitals or clinics. 535 children have been consecutively enrolled in the cohort during the years from 2011 to 2020. Patients with previous surgery of the knee, concomitant posterior cruciate ligament injury, or fractures were excluded for this study.

The children were asked to complete Pedi-IKDC before surgery and at 1 year follow-up. Information about surgical technique, concomitant injuries, and the treatment of these was registered prospectively in Copenhagen and acquired retrospectively from the medical files in Aarhus. For this study, all information was cross checked with the medical files at both centers. Data were registered according to a permission from the Danish Data Protection Agency (jr. nr. 2012-58-0004). The ethical committee stated that a specific permission was not necessary, as there was no intervention (H-20050959).

2.2 | The questionnaire

A Danish version of the Pedi-IKDC, translated from the original US English version by forward-backward translation, was published in 2016.¹⁰

The questionnaire consists of 21 items, of which one does not count in the score. There are two subscales: "Symptoms", comprising 9 items and "Sports activities", comprising 12 items. There are varying numbers of possible answers: Fifteen items have a 5-point Likert scale, two are dichotomous (yes/no), and four have a Visual Analogue Scale from 0 to 10. Raw scores from both subscales are aggregated, and the resulting total score is transformed into percentages of the maximum possible score.¹⁶

2.3 | Statistical analyses

2.3.1 | Structural validity

The structural validity of a questionnaire is a part of its construct validity, which describes whether the scale

validly measures the proposed construct (e.g., activities of daily living). With MTT models this includes analyses of response patterns for each item, in contrast to CTT methods which compare scores of (sub)scales only. This allows MTT models to identify specific items that do not fit well with the hypothesized structure of the questionnaire, indicating that the information these items contain is not reflected well by the scale scores. If such items are kept in the questionnaire, the precision of it is challenged due to measurement error and/or bias from the misfitting items. In that case, items may be altered or removed to obtain better measurement properties and more precise measurements. However, if misfitting items have been developed by methods that ensure their content validity, they should not necessarily be removed, but can be retained and reported as single items. Further, MTT models allow for assessment of dimensionality, revealing whether items placed in the same subscale actually measure the same underlying construct.¹⁵ Also, MTT models can confirm if scores reflect a value on an interval scale level, and not just data observed and reported on an ordinal scale level.¹³ This ensures that a difference in scores reflects the same real difference throughout the whole scale (e.g., the magnitude of an increase in pain from two to four is equal to an increase from five to seven). Dimensionality and scale level are both imperative when scale scores are interpreted, and they cannot be tested for with CTT methods.^{13,15}

By the fit of MTT models (CFA and IRT), we evaluated the total scale as a measurement of a single unidimensional construct (i.e., one total score; Figure 1, left panel), as this is how the questionnaire was described by the developers. Second, we evaluated the two subscales separately, as measurements of a two-factor construct divided in “Symptoms” and “Sports activities” (Figure 1, middle panel). Lastly, we applied bifactor models to evaluate whether the two subscales could adequately represent one common construct although with unique separate factors, also known as essential unidimensionality (Figure 1, right panel). We also evaluated the fit of these three measurement models in extended models taking local dependence into account (Figure 2).

CFA model fit was evaluated using the chi-squared statistic. This is the most substantive analysis, and p -values <0.05 indicate that the model does not fit the data. For completeness, the three indices of close fit (the root mean square error of approximation (RMSEA), the comparative fit index (CFI), and the Tucker-Lewis index (TLI)) are also reported. For RMSEA, values <0.06 are considered acceptable, and for CFI and TLI values above 0.95 are taken to indicate good fit.¹⁷ The chi-squared statistic is sensitive to sample size and is not suitable for model fit evaluation in large samples. Although the sample size was moderate, to

ensure that this phenomenon was accounted for analyses of smaller random subsamples ($N=100$) were included in the sensitivity analyses.

Since each MTT model has its own strengths and limitations, combining them can provide a more complete interpretation of data. Therefore, to achieve a more comprehensive assessment, IRT analysis was used in conjunction with CFA. This was done in order to confirm the CFA results and because IRT evaluates item fit. For the IRT analysis we used the graded response model. IRT item fit was evaluated using the generalized versions of Orlando and Thissen S-X2 item fit statistics¹⁸ and overall IRT model fit was evaluated using the C2 variant¹⁹ of the M2 fit statistic.²⁰ For the analysis of item fit, p -values <0.05 indicate that the specific item does not fit the data (i.e., that the observed PROM data differ significantly from the model expectations).

Data from both time points were included in the analyses and only configural invariance models were looked at. CFA analysis for ordered items was performed using the R package lavaan²¹ (version 0.6–15), and IRT analyses used the R package mirt²² (version 1.37.1) under R version 4.2.1.

We evaluated structural coefficient bias²³ by simulating Pedi-IKDC with a bifactor using the estimated parameters, but including an additional criterion variable with a pre-specified correlation with the general factor. The pre-specified values (chosen here as 0.0, 0.2, 0.4, and 0.6, respectively) are then compared to estimated values when the Pedi-IKDC is fitted to the simulated data as a unidimensional construct.

2.4 | Reliability

Reliability relates to whether the instrument is free from random measurement error. For unidimensional scales this is often reported as internal consistency by Cronbach's alpha. For multidimensional scales, it can be reported as McDonalds coefficient omega.²⁴ Coefficient omega applies to both unidimensional and multidimensional scales and is thus reported here. Values >0.7 are considered adequate and >0.8 considered good.

2.5 | Responsiveness

Responsiveness is the ability of an instrument to detect a true clinical change over time. A distribution-based approach, such as the standardized response mean, is the most frequently used.²⁵ Standardized response means (SRM) and 95% confidence intervals were computed using estimated person locations (factor scores) derived from

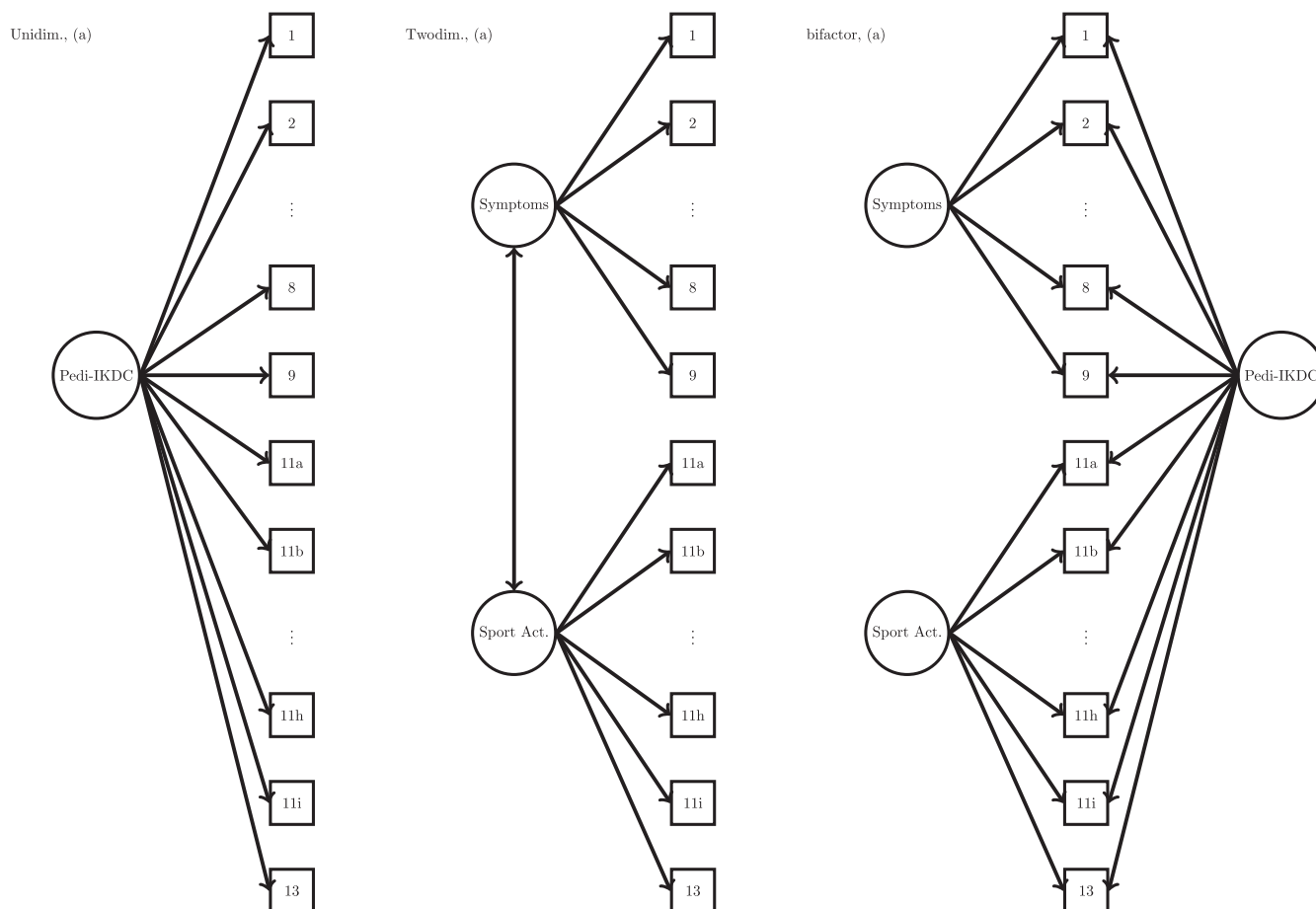


FIGURE 1 Factor structure for the CFA models for the Pedi-IKDC data. Left panel: the original unidimensional CFA model with all items proposed to reflect one unique underlying factor. Middle panel: the two-factor CFA model with items of each subscale reflecting two separate factors. Right panel: the bifactor CFA model with the two subscales sharing one common construct, but with separate factor loadings present (essential unidimensionality). Numbers represent the items in the subscale.

the fitted IRT models, and reported based on the subsample with responses at baseline and follow-up.

3 | RESULTS

3.1 | Patients

Pedi-IKDC data were available from 372 patients (Table 1). Of these, 22 patients were 16 years of age or older at baseline. In accordance with the definition of children by the Danish National Board of Health, we regarded <16 years as a pragmatic cutoff age, and excluded these patients regardless of physical morphology. 30 patients with previous surgery of the knee, concomitant posterior cruciate ligament injury, or fractures were also excluded. For one patient, data of concomitant injuries were not available. Therefore, PROM data from 319 patients were included. For the psychometric validation we included 156 responses at baseline (before surgery) and 269 responses at follow-up (106 respondents with responses at both time

points were included in the analyses of responsiveness). There were no missing values for the Pedi-IKDC items.

3.2 | Structural validity of the total Pedi-IKDC and of the Pedi-IKDC subscales

The total Pedi-IKDC scale did not fit the original unidimensional one-factor CFA model (Figure 1, panel (a)). This was evident in data from both time points (Table 2). A two-factor CFA model (Figure 1, panel (b)) improved close fit indices (RMSEA, CFI, TLI) slightly at both time points (Table 2), as did a bi-factor CFA model (Figure 1, panel (c)). Even if these indices improved slightly, the model fit was still poor. Thus, our analyses failed to confirm validity of the instrument. In the sensitivity analyses using smaller random subsamples ($N=100$) model fit was also rejected (results not shown). This was also evident from the tests of item fit (Table 3), where 16 out of 40 (40%) test statistics indicated statistically significant misfit. Six items showed misfit for both time points.

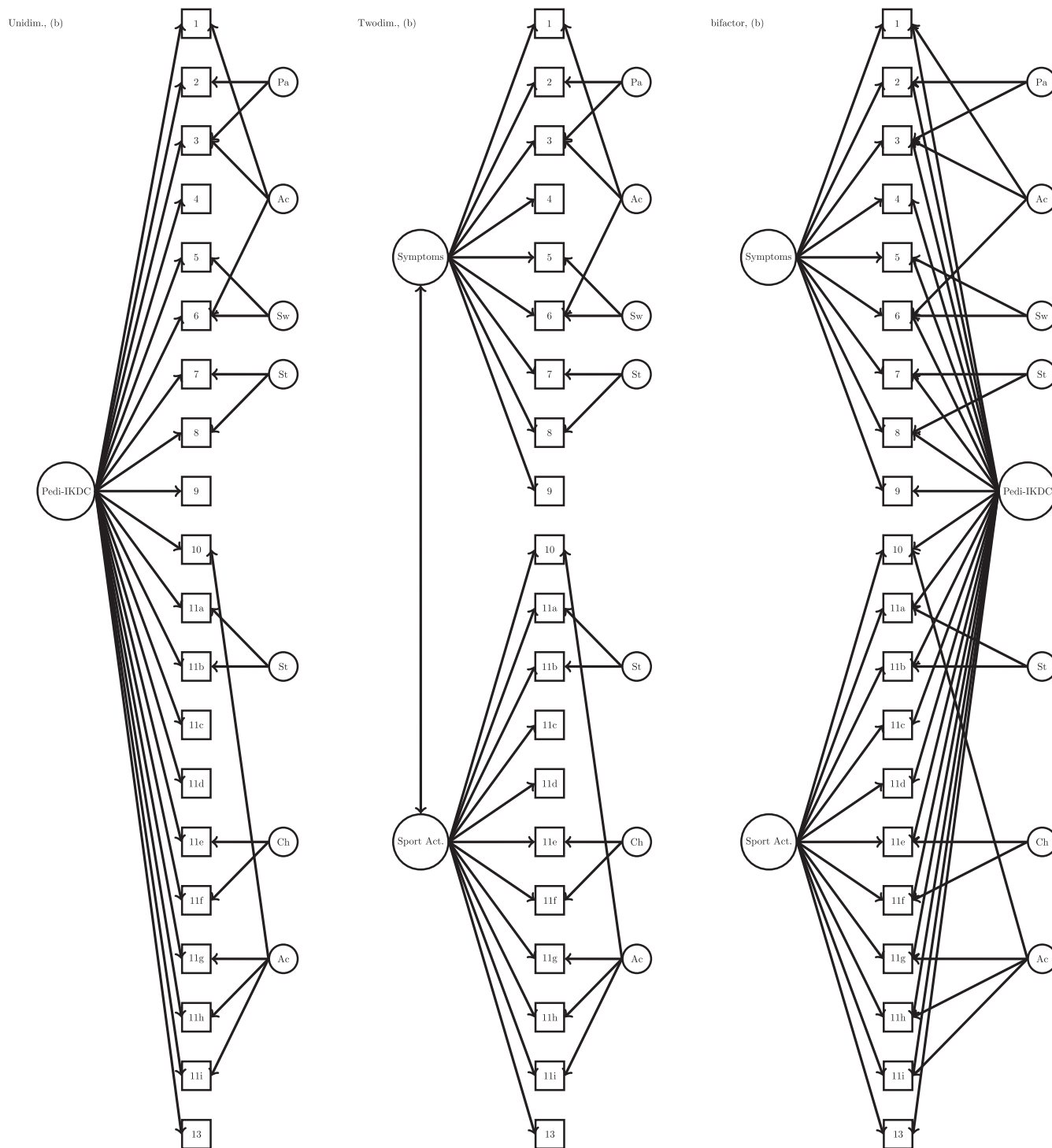


FIGURE 2 Factor structure for the CFA models for the Pedi-IKDC data. Left panel: the original unidimensional CFA model with all items proposed to reflect one unique underlying factor. Middle panel: the two-factor CFA model with items of each subscale reflecting two separate factors. Right panel: the bifactor CFA model with the two subscales sharing one common construct, but with separate factor loadings present (essential unidimensionality). Numbers represent the items in the subscale. Ac, activity- related items; Ch, chair; Pa, pain; S, symptoms subscale; SP, sports activities subscale; St, stairs; Sw, swelling.

The overall fit of CFA and IRT models yielded similar results (Table 2), as did the evaluation of the item fit (Table 3). The impact of the misfit is evident in the structural coefficient bias in data simulated using correlations

0.2, 0.4, and 0.6, respectively: estimated values were 0.238, 0.462, and 0.664 before surgery and 0.219, 0.430, and 0.627 1 year after surgery. Thus, before surgery the magnitude of bias was 19%, 16%, and 11% and 1 year after surgery the

TABLE 1 Patient characteristics.

Cohort, <i>n</i>	535
PROM data available	372
Excluded, <i>n</i> (%)	53 (14.1)
Included	319
Females (% of cohort)	157 (49.2)
Males (% of cohort)	162 (50.8)
Age	
Mean	13.5
Range	9–15
Injury, <i>n</i> (% of cohort)	
ACL	195 (61.1)
ACL + meniscal injury	99 (31.0)
ACL + cartilage injury	8 (2.5)
ACL + collateral ligament injury	1 (0.3)
ACL + meniscal + cartilage injuries	11 (3.5)
ACL + meniscal + collateral ligament injury	4 (1.3)
ACL + meniscal + cartilage + collateral ligament injuries	1 (0.3)

magnitude of bias was 10%, 8%, and 5%. Standardized factor loadings for the CFA model (Appendix S1) and polychoric correlation matrices (Appendix S2) are reported in the supplementary material.

3.3 | Reliability

McDonalds coefficient omega was 0.866 before surgery and 0.919 at 1-year follow-up.

3.4 | Responsiveness

The original total score was able to detect changes over time with an SRM value of 1.66 (95% CI: 1.46–1.88).

4 | DISCUSSION

This study showed that Pedi-IKDC does not exhibit adequate structural validity for children with ACL injury when assessed by CFA and IRT models. The original construct of Pedi-IKDC as a unidimensional scale could not be confirmed to fit the statistical models, and although bifactor models fitted data better, the model fit was still inadequate. This was confirmed by the 16 out of 40 (40%) item fit test statistics indicating statistically significant misfit. It is very unlikely that these are all Type I errors. Furthermore, evidence of local response dependence was

TABLE 2 Evaluation of CFA and IRT fit for the “Symptoms” and “Sports activities” subscales. Results shown without (A) and with (B) local dependence taking into account (by adding the bifactors activity, pain, swelling, chair, and stairs (Figure 2)).

Time	Model	CFA				IRT			
		χ^2	df	<i>p</i>	RMSEA	CFI	TLI	M2	<i>p</i>
Before surgery	(a) Unidimensional	765.5	170	<0.0001	0.150	0.821	0.800	736.2	<0.0001
	Two-factor	702.5	169	<0.0001	0.142	0.890	0.820	2199.7	<0.0001
	Bifactor	443.5	150	<0.0001	0.112	0.912	0.888	654.1	<0.0001
	(b) Unidimensional	462.0	163	<0.0001	0.109	0.910	0.895	335.1	<0.0001
	Two-factor	455.6	162	<0.0001	0.108	0.912	0.897	404.3	<0.0001
	Bifactor	338.9	143	<0.0001	0.094	0.941	0.922	232.1	<0.0001
1-year follow-up	(a) Unidimensional	716.3	170	<0.0001	0.109	0.919	0.910	878.6	<0.0001
	Two-factor	676.4	169	<0.0001	0.106	0.925	0.915	4135.9	<0.0001
	Bifactor	413.7	150	<0.0001	0.081	0.961	0.950	589.6	<0.0001
	(b) Unidimensional	538.2	163	<0.0001	0.093	0.944	0.935	486.4	<0.0001
	Two-factor	541.3	162	<0.0001	0.093	0.944	0.934	783.8	<0.0001
	Bifactor	279.6	132	<0.0001	0.064	0.978	0.969	337.7	<0.0001

TABLE 3 Item fit evaluation for bifactor IRT models taking local dependence into account (by adding the bifactors activity, pain, swelling, chair, and stairs (Figure 2)). Bold type indicate lack of fit to the model ($p < 0.05$).

Item	Before surgery			1-year follow-up		
	S-X ²	df	p	S-X ²	df	p
IKDC_1	41.9	27	0.0340	30.8	18	0.0300
IKDC_2	45.8	38	0.1809	18.9	9	0.0259
IKDC_3	40.9	29	0.0708	21.1	10	0.0206
IKDC_4	30.3	27	0.3020	11.0	13	0.6147
IKDC_5	31.1	31	0.4633	13.8	7	0.0548
IKDC_6	32.7	20	0.0366	26.6	16	0.0462
IKDC_7	14.9	15	0.4614	19.9	14	0.1349
IKDC_8	11.3	16	0.7882	21.7	14	0.0859
IKDC_9	28.1	28	0.4586	15.3	17	0.5720
IKDC_10	35.8	22	0.0319	36.3	23	0.0385
IKDC_11a	33.1	25	0.1297	17.9	10	0.0563
IKDC_11b	50.9	27	0.0036	6.6	10	0.7643
IKDC_11c	46.4	48	0.5373	29.6	28	0.3846
IKDC_11d	51.4	47	0.3071	44.7	33	0.0836
IKDC_11e	22.1	13	0.0540	12.7	11	0.3148
IKDC_11f	19.8	19	0.4089	7.7	4	0.1013
IKDC_11g	43.6	33	0.1020	30.9	18	0.0299
IKDC_11h	66.4	35	0.0011	35.7	19	0.0115
IKDC_11i	54.3	34	0.0149	115.1	29	<0.0001
IKDC_13	73.5	46	0.0062	91.7	32	<0.0001

Abbreviations: IKDC_1, activity level; IKDC_10, activity in sports; IKDC_11a, walk up stairs; IKDC_11b, walk down stairs; IKDC_11c, kneeling; IKDC_11d, squatting; IKDC_11e, sit in a chair; IKDC_11f, rising from chair; IKDC_11g, running; IKDC_11h, jump and land; IKDC_11i, accelerating/decelerating; IKDC_13, knee function today; IKDC_2, pain over time; IKDC_3, pain today; IKDC_4, bending and moving; IKDC_5, swelling; IKDC_6, swelling during activities; IKDC_7, true locking; IKDC_8, sensation of locking; IKDC_9, giving way during activities.

disclosed which means that the previously published estimates of reliability of the instrument may be incorrect, yielding a false impression of its precision.²⁶ Considering that the questionnaire also possesses a low degree of content validity for these children, due to the way it was developed, data obtained by Pedi-IKDC should be interpreted with caution. In particular, the risk of Type II errors in clinical trials reporting Pedi-IKDC data are high.

4.1 | The clinical consequences

The MTT models revealed Pedi-IKDC to be a multidimensional scale, meaning that it does not function as one construct with one aggregated score. As a consequence, sum-scores make little sense. Inadequate construct validity hampers the ability of a questionnaire to detect changes over time,²⁷ and in clinical trials, a significant difference between treatment groups is found twice as often in studies that use the most adequate PROM compared to studies using a less adequate PROM.²⁸

This study contrasts previous conclusions of Pedi-IKDC to be valid, reliable, and responsive.^{9–12} However, our findings cannot be directly compared to previous findings derived using CTT methods. Since MTT models are the most suitable methods to analyze construct validity,¹⁵ it is not uncommon that results obtained by CTT methods are rejected by MTT-analyses, when the prerequisites to perform a CTT analysis are not present. For Pedi-IKDC, for instance, it is a prerequisite for calculation of reliability by CTT that the scale is unidimensional, which the present study shows is not the case.

4.2 | Potential reasons for inadequate structural validity

The adult IKDC served as template for Pedi-IKDC.^{6,7} IKDC was developed by an expert panel without involvement of patients, hence IKDC has no patient-proven content validity.^{16,29–31} This is an important factor leading to inadequate measurement properties,

which is pointed out in the 2018 International Olympic Committee consensus statement on pediatric ACL injuries.³² Even though children were interviewed about their difficulties related to answering the adult items, it was not explored whether they found items relevant, and whether additional items were needed to cover all key aspects of the condition.

The lack of patient involvement in the development process may explain why the scoring of Pedi-IKDC is problematic. Children find it difficult to interpret and answer questions with too many response options.³³ In Pedi-IKDC, 18 of 20 items contain five or eleven response options, which might explain some of the irregular response patterns that MTT models detect. This was in fact pointed out by the developers of Pedi-IKDC,⁶ and it was suggested to convert the 11-point VAS-scales into Likert scales.⁷ Wording was also described as confusing, for example, Item 4, “How hard has it been to move or bend your injured knee”.⁵ This is an aggregate of “how hard has it been to move your knee”, and “how hard has it been to bend your knee”. However, the answers to these questions are not necessarily identical. Moreover, it is a double-barreled question, containing one question about movement and one specifically about bending. Such questions are confusing and should be avoided, or split into two separate items.³⁴

The scoring of the scale has some inherent potential problems: the maximum points per item varies between 1 point (e.g., Item 7 and 8) and 10 points (e.g., Item 2 and 3), meaning that items can weigh up to 10-fold differently without obvious justification. The two defined subscales, “Symptoms” and “Sports activities”, contain maximum scores of 42 and 50, respectively, meaning that “Sports activities” contributes more to the overall score and is thus considered more important. Whether this unequal weigh is appropriate has not been mathematically assessed.

All of these potential problems negatively affect the measurement properties and explain why Pedi-IKDC is inadequate in its current form. It is uncertain to which degree subscale scores represent how children are doing, and changes in Pedi-IKDC—for instance in randomized studies or cohorts—is at high risk of errors, especially, but not exclusively, Type II errors. Therefore, the limitations of the score should be discussed when results are interpreted.

4.3 | Strengths and limitations

The inclusion of two different MTT models, and the fact that the models reach the same conclusions, is a

strength of this study. Further, the nationwide cohort minimizes the risk of selection bias. A limitation is that the fit criteria used for the close fit indices rely on ML estimation for continuous items, whereas our analysis is CFA analysis for ordered items, which may influence the values of RMSEA, CFI, and TLI. The results of this study relate to how Pedi-IKDC functions in a cohort of children with an ACL injury, and are not necessarily valid in other populations.

5 | CONCLUSION

Pedi-IKDC did not exhibit adequate fit to the robust statistical models CFA and IRT analysis in a nationwide, consecutive cohort of children with ACL injury. The lack of fit could be a result of several problematic issues, in particular inadequate transformation from the adult IKDC, flawed comprehensibility, lack of relevance, and lack of coverage. We suggest that conclusions from clinical studies based on Pedi-IKDC scores are interpreted with caution.

6 | PERSPECTIVE

Pedi-IKDC is one of only two pediatric PROMs for assessing knee injuries, and is the most frequently used for evaluating injuries to the ACL in the pediatric population. Thus, data from Pedi-IKDC is likely to contribute to health care strategies and clinical guidelines within this field. The insufficiencies of Pedi-IKDC described in this study should lead to an increased awareness of measurement errors in trials using Pedi-IKDC as outcome before conclusions are put into practice. As there is currently no valid PROM for this large and important patient population, the development of a new PROM targeting children with ACL deficiency is essential.

FUNDING INFORMATION

The authors declare no specific grant for this research.

CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

How to cite this article: Hansen CF, Madsen MØ, Warming S, et al. Pedi-IKDC exhibits questionable measurement properties in a cohort of pediatric patients with ACL rupture. *Scand J Med Sci Sports*. 2023;00:1-10. doi:[10.1111/sms.14413](https://doi.org/10.1111/sms.14413)